Assignment 2

This exercise sheet is mainly devoted to linear systems and the condition number of a matrix.

Graded exercise

Exercise 10 (Condition number of the finite difference matrix)
We consider the following tridiagonal matrix $A_N \in \mathbb{R}^{N\times N}$:

$$A_N = \begin{pmatrix} 2 & -1 & 0 & 0 & \dots & 0 \\ -1 & 2 & -1 & 0 & \dots & 0 \\ 0 & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & -1 & 2 & -1 & 0 \\ 0 & \dots & 0 & -1 & 2 & -1 \\ 0 & \dots & \dots & 0 & -1 & 2 \end{pmatrix}. \tag{1}$$

This matrix is associated to the numerical solution of boundary value problems (it is the matrix which appears in the motivational example given in class). We are interested in studying, theoretically and numerically, the behavior of the condition number of A_N as $N \to \infty$. In the framework of the numerical solution of boundary value problems, $N \to \infty$ corresponds to the case that the solution gets closer and closer to the exact one.

(a) Write a PYTHON script that, for each $N = 2^k$ with $k = 4, \dots, 12$, allocates the matrix A_N as in (1) and computes its condition number $\kappa_2(A_N)$. Furthermore, plot the obtained condition numbers versus the size N of the matrix.

Hint: To allocate a tridiagonal matrix, the function `numpy.diag` is of great help.

Hint: To compute the condition numbers, you can use the built-in function `numpy.linalg.cond`.

(b) From the previous task, you should observe a polynomial increase of the condition number with N: $\kappa_2(A_N) \approx CN^{\alpha}$, for some $\alpha \geq 1$ and $C > 0$ a constant. Add a few lines of code to the script for (a) to estimate α from the data $(N, \kappa_2(A_N))$ and print the result on screen. Which value do you observe?

Hint: We can rewrite $\kappa_2(A_N) \approx CN^\alpha$ as $\log(\kappa_2(A_N)) \approx \alpha \log(N) + \log(C)$. The latter is a *first order polynomial*, i.e. a line, with respect to $\log(N)$, and the highest order coefficient, i.e. the slope of the line, is exactly the power α we are looking for. This means that taking the highest order coefficient in a linear fit of $\log(\kappa_2(A_N))$ versus $\log(N)$ gives us an estimate for α. To perform this, you can use either the function `numpy.polyfit` or the more recent function `numpy.polynomial.Polynomial.fit`. Note that the two functions use different conventions: with `numpy.polynomial.Polynomial.fit`, to obtain the coefficients you need to first use `convert` and the coefficients are ordered from lowest to highest order in monomial degree; with `numpy.polyfit` instead, you obtain the coefficients ordered from highest to lowest order in monomial degree.

(c) We now aim at justifying theoretically the behavior observed numerically. Since A_N is a real, symmetrix matrix for every N, we have $\kappa_2(A_N) = \frac{|\lambda_{\max}(A_N)|}{|\lambda_{\min}(A_N)|}$ (see also Exercise 11). Here $\lambda_{\max}(A_N)$ and $\lambda_{\min}(A_N)$ denote, respectively, the maximum and minimum eigenvalues of A_N. It can be shown that the eigenvalues of A_N are given by

$$\lambda_n = 4\sin^2\left(\frac{n\pi}{2(N+1)}\right), \quad \text{for } n = 1, \dots, N.$$

Use this information to estimate theoretically the behavior of $\kappa_2(A_N)$ as $N \to \infty$ and to justify the rate of increase α that you observed numerically in task (b).

Other exercises

Exercise 11 (Spectral condition number)

In this exercise, we focus on better understanding the claim that, if a matrix A is real, symmetric and non-singular, then $\kappa_2(A) = \frac{|\lambda_{\max}(A)|}{|\lambda_{\min}(A)|}$, where $\lambda_{\max}(A)$ and $\lambda_{\min}(A)$ denote, respectively, the maximum and minimum eigenvalues of A in absolute value[1]. Then we generalize the result to non-singular matrices (not necessarily diagonalizable ones).

(a) Let $A \in \mathbb{R}^{m\times m}$ be a symmetric matrix. Show that $\|A\|_2 = |\lambda_{\max}(A)|$, where $\|\cdot\|_2$ denotes the norm induced by the Euclidean vector norm.

Hint: Use the definition $\|A\|_2 = \sup_{v\neq 0} \frac{\|Av\|_2}{\|v\|_2}$ and the fact that, since A is real and symmetric, its (normalized) eigenvectors form an orthonormal basis of $\mathbb{R}^m$.

(b) Use the definition of the condition number to conclude that, for A as in the previous task and non-singular, $\kappa_2(A) = \frac{|\lambda_{\max}(A)|}{|\lambda_{\min}(A)|}$.

[1]Note that, differently from what was written on the blackboard in class, the matrix is not necessarily positive definite, hence the absolute values for the eigenvalues.

(c) Now we only assume that $A \in \mathbb{R}^{m\times m}$ is non-singular. Show that $\|A\|_2 = \sqrt{|\lambda_{\max}(A^\top A)|}$, where $\lambda_{\max}(A^\top A)$ is the maximum eigenvalue of $A^\top A$ (note that $A^\top A$ is symmetric hence diagonalizable).

(d) For a matrix $A \in \mathbb{R}^{m\times m}$, its singular values are defined as $\sigma_i(A) := \sqrt{|\lambda_i(A^\top A)|}$, for $i = 1, \ldots, m$ and $\lambda_i(A^\top A)$ the i-th eigenvalue of $A^\top A$, ordered from largest to smallest absolute value. Use the result from the previous task to show that, for a generic non-singular matrix $A \in \mathbb{R}^{m\times m}$, $\kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}$, where the notation $\sigma_{\max}(A) = \sigma_1(A)$ and $\sigma_{\min}(A) = \sigma_m(A)$ has been used.

Exercise 12 (Condition number of a matrix)
Let $\varepsilon \in \mathbb{R}$ be such that $0 < |\varepsilon| \le 2$. Compute the condition number $\kappa_\infty(A)$ of the matrix

$$A = \begin{pmatrix} 1 & 0 \\ 1 & \varepsilon \end{pmatrix}$$

with respect to the matrix norm $\|A\|_\infty := \max_{1\le i\le m} \sum_{j=1}^m |a_{ij}|$ (here $m = 2$). What's the behavior of $\kappa_\infty(A)$ as $\varepsilon \to 0$?